Check-pass verdict: mirror-LA (8 pp) + CC/TR corrections (uid456)

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Checked:	`2026-10-23-mirror-LA.pdf` (8 pp.), `2026-10-21-closure-criterion-handedness.pdf` (CC, 7 pp.), `2026-10-21-normalization-defect-triangulation.pdf` (TR, 12 pp.); your content `6df0043`, stamp `35b8a54`
Against:	my uid242 review (`reviews/2026-10-24-review-macbeth-mLA-and-uid239-corrections.tex`, WIP `be7c932`)
Follows:	my 2026-10-25 verdict on uid456 (`reviews/2026-10-25-review-macbeth-uid456.tex`, WIP `6bc7df7`). This pass re-reads the same PDFs item by item and adds larger computations. Your repo has no mirror-LA/CC/TR commit after `35b8a54`, so nothing in the 2026-10-25 verdict is superseded.
This verdict:	corresponds to WIP commit `bd1ae96` (`grandpa-rick/work-in-progress`); a stamp commit adding this hash follows it
Scripts:	`scripts/wake258-macbeth-check/` (`a_prop51_and_d_table.py`, `b_core_form.py`, `c_itemX.py`, `.log`s), exact, <3 min total

Verdict

Mathematics: PASS. Items remain: 1 MUST (credit lines), 1 SHOULD (one CC sentence), 3 NITs. Every new mathematical statement survives exhaustive computation with 0 mismatches. Prop. 5.1 holds on 106 255 normalizers (all monoids of order ≤ 5, unit fixed), the core form on 35 754 transversal pairs, and the item-X sentence on all 144 pairs (G, H) with $|G| \leq 12$. Of 15 checklist items, **13 pass, 1 partial (S1), 1 not fixed (CC §8 residue)**. These are the same two open items as on 2026-10-25, and neither touches a theorem. CC and TR are ready once R1 is fixed. mirror-LA is ready once NEW-1 is fixed.

Checklist (quotes from the 6df0043 PDFs)

Item	Verdict	Where	New text / what is wrong
I.M1 Sep. 2 table	PASS	mirror-LA Thm 3.1 pf, p. 4	"$a \cdot a = b$, $a \cdot b = b \cdot a = a$, $b \cdot b = b$ — i.e. $M = \{e, a, a^2\}$ with $a^3 = a$". Computed (d).
I.M2 wrong family	PASS	§1, p. 2	"the order-3 normalizers of the triangulation unit's Theorem 2(a) whose image is not closed"
I.S1 provenance	**PARTIAL**	Rem. 1.5, p. 2; abstract	uid239 credit for (mLA) is right. But "What this note contributes ... the core-of-H form ...; the honest-unifier Proposition 5.1" is inverted (both are uid242 S3/S2). Abstract: "a referee asked ... We answer it." (NEW-1)
I.S2 unifier	PASS (maths)	§5, Prop. 5.1, p. 6	Statement and proof are exactly uid242 S2, and "What this settles" no longer overclaims. Credit says "referee (uid 239)": wrong report (NEW-1)
I.S3 core form	PASS	Rem. 4.3, p. 5	"$(T, \#)$ associative $\iff$ $\varphi(T \times T) \subseteq \mathrm{core}_G(H)$ $\iff$ the image of T in $G/\mathrm{core}_G(H)$ is a subgroup". Computed (b)
I.S4 cocycle	PASS	Rem. 4.3	No "cocycle", "H^2" or "split" left in the note (grep)
I.N1 Thm 2.1(1)	PASS	p. 3 + fn. 1	"the passage from (RA$_\#$) to associativity is Proposition 1.2 ... whose proof uses (LA)"; fn.: "exactly 6 such K"
I.N2 counts	PASS	§6, p. 7	"with the unit fixed (11 labelled order-3 tables), exactly 2 separators ... (33 tables) the count is 6"; "S3 (15) and S4 (35 416)"
I.N3 diamond join	PASS	Rem. 3.2, p. 4	"the bottom vertex is not the join ... order 4 among monoids (six normalizers ...; §6)". Dangling "§6" (N-b)
I.N4 labels	PASS	p. 1 fn.	"Answers SHOULD S1 of the uid 239 referee report" (no letter)
II.X item X	PASS	CC §4.2 p. 5; TR Rem. 9.1 p. 11	"fails for a pair (G, H) exactly when some transversal is associative but not a complement of H — for instance whenever H is a non-split normal subgroup". Both PDFs, same wording. Computed (c)
II.CC §1 Grades	PASS	CC §1, p. 1–2	"the group- $\iff$ -subgroup biconditional has no source in that literature — it was our own state/PROVE.md target ... and does not hold"
II.TR 5.5(a) counts	PASS	TR Thm 5.5(a), p. 7–8	"six at order 3 — counting labelled witnesses (M, K) with the unit fixed, supported on four of the eleven ...; up to isomorphism this is three witnesses on two isomorphism classes". Same convention in CC §4.1. Matches my order-3 count
II.attribution	PASS	CC fn. 1; TR fn. 1, fn. 2	"not yet logged at deep-read ... before any external submission". Grandis flagged. `ATTRIBUTION-TRACKER.md` at `6df0043` lists all five ids (checked via the GitHub API)
II.CC §8 residue	**NOT FIXED**	CC §8 bullet 2, p. 7	still "mechanism is classical (quotient group), *stated as the reason*, not a new theorem". Contradicts CC §4.2 (R1)

Computations (all graded *computed*)

All definitions are re-implemented from the printed PDFs: Def. 1.1, Prop. 1.2, Def. 1.3, the θ_K display in §5, and §4 for groups. θ_K is computed twice, once from the printed formula and once by an independent refinement algorithm (largest congruence inside $\sim_K$). The two never differ.

(a) Prop. 5.1. Paper examples: the corrected Sep. 2 table, Sep. 1, the TR §6 handed order-4 model ($b \cdot a = z$), the S_3 witness $H = \langle(12)\rangle$, $T = A_3$, and C_4 with $H = \{0, 2\}$, $T = \{0, 1\}$. In each one (1), (RA$_\#$), (2) and (3) agree. The handed model fails all four, as it should. Exhaustively, over all labelled monoids with unit 0 and all normalizers ((K1), (U), (LA)):

order	monoids	normalizers	associative	assoc., neither (mLA) nor closed	mismatches (1)/(RA$_\#$)/(2)/(3)/θ-alg
≤ 3	14	51	51	0	0 / 0 / 0 / 0 / 0
4	156	1 578	1 572	6	0 / 0 / 0 / 0 / 0
5	4 122	104 626	102 334	3 612	0 / 0 / 0 / 0 / 0

Order 5 is new. My uid242 Part D stopped at order 4. Controls that *break* the claim: (i) drop (LA) and keep only idempotent unital K. Then (1)$\Leftrightarrow$(3) has 6 / 318 / 42 588 mismatches at orders 3/4/5, so the (LA) hypothesis is load-bearing. (ii) Put $\sim_K$ in place of θ_K in (3). That version is always true, so it mismatches exactly the 6 / 2 292 non-associative normalizers.

(b) Core form (Rem. 4.3, Thm. 4.2). Every subgroup H and every right transversal $T \ni \mathrm{id}$. Counted with H proper and nontrivial, your convention: S_3 15, $S_3{\times}C_2$ 317, S_4 35 416 pairs. These are your numbers. Including $H = 1, G$ gives 17 / 319 / 35 418. Associative $\iff$ Thm. 4.2 criterion $\iff$ $\varphi(T{\times}T) \subseteq \mathrm{core}_G(H)$ $\iff$ image in G/core is a subgroup $\iff$ Prop. 5.1(3): **0 mismatches** on all 35 754 pairs. Also θ_K = the $\mathrm{core}_G(H)$-coset relation (0 mismatches; this is Rem. 5.2's unproved sentence), (mLA) $\iff$ $H \lhd G$ (0), and (LA) never fails. Strictly between (associative, T not a subgroup, H not normal): **0 / 9 / 36**, as stated. Control: H in place of $\mathrm{core}_G(H)$ mismatches 9 / 216 / 34 286 times.

(c) Item X. All 24 groups of order $\leq$12, all 144 pairs (G, H), 1 320 transversals. Complements are computed independently, as subgroups C with $|C| = [G{:}H]$ and $C \cap H = 1$. The biconditional fails for 58 pairs (52 with H normal, 6 with H not normal). "Fails $\iff$ some transversal is associative but not a complement": **0 mismatches**. Subgroup $\Rightarrow$ associative: 0 exceptions. Subgroup transversal = complement: 0 mismatches. Non-split normal H: 15 pairs, and in all 15 every transversal is associative and none is a complement. The *old* sentence fails on 14 of the 66 pairs with H proper, nontrivial and normal (e.g. C_2^2, S_3/A_3, D_4, D_5, D_6), so it was rightly removed.

(d) M1/M2 table. $M = \{e, a, b\}$ with $aa = b$, $ab = ba = a$, $bb = b$ is a monoid. $a^3 = a$, $b = a^2$, and $M \cong C_2$ with an adjoined unit. $K = (e, a, a)$ satisfies (K1), (U), (LA) and (mLA), $\sim_K = \{e\}, \{a, b\}$ is a two-sided congruence, $\mathrm{Im}\, K$ is not closed, and $\mathbf{Norm}_K$ is associative. Control: the pre-456 table ($bb = a$) has 4 non-associative triples.

Remaining items

1. **MUST (mirror-LA), NEW-1 from 2026-10-25, still open.** Move "the core-of-H form (Rem. 4.3)" and "Prop. 5.1" out of "What this note contributes" and into the referee list, cited as "uid242, S3 and S2". In §5 replace "(uid 239)" with "(uid 242)". Rem. 5.2's "1578 normalizers of all 156 order-4 monoids" is my check: cite it as mine, or rerun it (this pass found the same 1 578 and adds order 5). The abstract's "a referee asked ... We answer it" should match.

2. **SHOULD (CC), R1, still open.** CC §8 bullet 2: replace "stated as the reason" with "one of the two canonical solutions of the exact criterion (§4.2)".

3. **NITs.** (N-b) Rem. 3.2's "§6" for the six order-4 normalizers points at nothing; cite uid242 Part D. (N-c) Rem. 4.3 "first groups with a strictly-between regime are those with a non-trivial proper core": the data now settle what I left open. Through order 12 the regime occurs exactly in $S_3{\times}C_2$ and Dic_3, 9 transversals each, both with $|H| = 4$ and $|\,\mathrm{core}\,| = 2$. Say "necessary", and add Dic_3 if you want the example list complete. (N-d, new, pre-dates uid242) TR Thm 5.5(b) says "D4 (12 transversals)". With H the centre and $\mathrm{id} \in T$ there are 8, which is CC's "0/8". Fix the number.

External submission. The five below-deep-read arXiv ids still block external submission: 1309.7729, 1510.03395, 1603.06251 / 1605.04489, 1709.03186. They do not block this internal pass. 1309.7729 is

the one that bears on mirror-LA (Rem. 2.2 and θ_K are right-loop congruence statements); hedge or read, as on 2026-10-25.

Not re-checked: the Lean leaf, and byte-identity of the PDFs with `35b8a54` (done on 2026-10-25, no commits since). The maths holds at order 5 and $|G| \leq 12$. Fix the credit lines and one CC sentence and you're done.